# Ajay Venkatesh

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## Skills

**Programming Languages:** Python, Java, Scala, R, SQL, C++, TypeScript, JavaScript (ES6)
**Big Data & Processing:** PySpark, Apache Spark, Pandas, NumPy, ETL/ELT Workflows, Distributed Data Pipelines, Airflow
**Databases:** SQL Server, MySQL, PostgreSQL, DynamoDB, MongoDB, Dataverse
**Cloud & Warehouses:** AWS (Redshift, S3, EC2, Lambda, Glue, CloudWatch, SageMaker, QuickSight, CDK), GCP (BigQuery), Azure (Data Lake, Function Apps)
**Tools:** Git, Docker, Kubernetes, Jupyter, Postman, Power BI, Visual Studio, Cursor
**AI/ML:** TensorFlow, LLM Fine-tuning, RAG (LangChain, FAISS), NLP, Deep Learning, RL.

## Experience

**Campus Mesh** **Jan 2026 – Present**
*Data Engineer* *New York, US*
- Developed scalable **Node.js microservices** powering real-time academic workflows for **10K+ users** across study groups, messaging, notifications, and profile management, backed by a **MongoDB** store.
- Designed a personalized **recommendation engine** using collaborative filtering on behavioral event data with **Python** feature pipelines, increasing engagement by **28%** and retention by **50%**.
- Engineered **LLM-based retrieval pipelines (RAG)** for CampusIQ, building ingestion/indexing workflows for structured data into an **embedding-based semantic search** store, improving accuracy by **30%**.

**Amazon** **May 2025 – Aug 2025**
*Software Development Engineer Intern* *Seattle, US*
- Architected a secure data console for Amazon fulfillment center teams to query image/video evidence across FCs; presented **system design** for approval and set up Role Based Access **(4+ roles)** via **OAuth, Midway SSO, HELIS**.
- Provisioned **serverless Infrastructure-as-Code** using **AWS CDK** with API Gateway, Lambda, **DynamoDB, S3**, IAM, **CloudWatch**, cutting provisioning time **70%** with **multi-stage, multi-region CI/CD**.
- Engineered distributed backend services in **Java** using Coral, Smithy, Dagger, CBOR; applied **low-level design patterns**, cutting execution time **35%** with unit/integration tests via **Mockito**.

**New York University (Novel AI Technologies, Inc.)** **Aug 2024 – Present**
*Data Engineer* *New York, US*
- Architected an **NSF-funded ($100K)** AI-powered health platform for **iOS** and **Android**, ingesting real-time wearable telemetry into cloud storage for downstream AI model inference on health metrics and food images.
- Engineered **ETL/ELT workflows** with **Python, PySpark, SQL**, transforming **NoSQL** event data into **PostgreSQL** with **concurrency control and locking**; processed **250K+** records and powered **AWS QuickSight** dashboards.
- Built a full-stack insurance analytics platform deployed via **Docker** containers on **EC2**, and implemented Stripe billing, RBAC, and row-level security, generating **$50K+** revenue across **6 clients**.
- Implemented **document/media processing pipelines** with **WebSocket real-time data streams** to streamline underwriting via a broker portal integrated with insurer platforms.

**PricewaterhouseCoopers (PwC)** **Aug 2021 – Aug 2024**
*Data Engineer* *Bengaluru, India*
- Developed a **distributed data processing backend** on **Microsoft Azure** integrating with **Microsoft CRM**, with **event-driven ingestion** and fault-tolerant retries over **SQL Server** stores.
- Deployed an **NLP-based OCR neural model** on Azure to automate structured data extraction from unstructured documents into downstream warehouses, reducing processing time by **75%**.
- Optimized batch ingestion with **parallel processing** for **100K-row** Excel feeds; reduced execution from **5 hours to 1 hour**.
- Built **Power BI** dashboards over datasets joined across multiple relational databases, improving reporting efficiency by **40%**.

## Projects

**Advanced Expense Tracker (Python Data Pipeline Application)** | ***Python, SQLite3, Pandas, Matplotlib, Tkinter***

- Architected a modular **Python application** supporting individual and group expense data with secure login, applying **OOP principles** and **MVC & DAO design patterns** over a **SQLite3** persistence layer.
- Implemented concurrent transaction handling and real-time cost aggregation using **Python threading** and locks to ensure thread-safe database writes; visualized spending trends with **Pandas**-backed analytics and Matplotlib charts.

**Flight Delay Prediction (Big Data)** | ***Hadoop HDFS, PySpark, Spark SQL, Spark MLlib, Seaborn, Matplotlib, Plotly***

- Built a distributed big data pipeline over **Hadoop HDFS** ingesting **2024 BTS US flight data**; used **PySpark** and **Spark SQL** for cleaning, feature engineering, and large-scale aggregations.
- Trained **Spark MLlib** classification models to predict delay severity and surfaced feature importance and delay distributions via **Seaborn, Matplotlib, and Plotly** visualizations.

## Education

**New York University** | *Master of Science in Computer Engineering* ***(GPA – 3.9/4)*** Sep 2024 – May 2026
- *Coursework:* Big Data, System Design, Machine Learning, Deep Learning, Data Structures

## Certifications

**AWS Cloud Practitioner**
Azure Data Fundamentals
Microsoft Solution Architect Expert
Power BI Data Analyst Associate
Microsoft Developer Associate
Stanford Machine Learning (Coursera)